MIL-C-52661B(ME)
27 September 1983
SUPERSEDING
MIL-C-52661A(ME)
25 June 1974

MILITARY SPECIFICATION

CONTAINERS, CARGO

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers 20-foot reusable cargo containers for the transportation and distribution of military supplies.

1.2 Classification. Containers shall be of the following types, as specified (see 6.2):

Type I - 8'-0" High ISO-1C without mechanical restraint system.
Type II - 8'-0" High ISO-1C with mechanical restraint system.
Type III - 8'-6" High ISO-1CC without mechanical restraint system.
Type IV - 8'-6" High ISO-1CC with mechanical restraint system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-530	- Plywood, Flat Panel.
TT-P-98	- Paint; Stencil, Flat.
VV-G-671	- Grease, Graphite.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8115

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-P-17549	- Plastic Laminates, Fibrous Glass Reinforced, Marine Structural.
MIL-C-46168	- Coating Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-83933	- Coating Compound, Bituminous Solvent Type, under Body, (For Motor Vehicles).

STANDARDS

MILITARY

MIL-STD-171	- Finishing of Metal and Wood Surfaces.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION

Bureau of Explosives.
BOE-6000-C-Hazardous Materials Regulations of the DOT.

Code of Federal Regulations.
CFR 49 - Transportation, Parts 450 and 451.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

DRAWINGS

ME

13219E0039 - Beam, Shoring.
13219E0080 - Wood Plank, Floor, Intermediate.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

MH5.3 - Identification and Marking of Cargo Containers.

International Standards Organization ISO/1161 - Specification of Corner Fittings for Series 1 Freight Containers.

International Standards Organization ISO/1496/I - Specifications and Testing of Series 1 Freight Containers.

International Standards Organization ISO/1496/VII - Specifications and Testing Part VII: Air Mode Containers Annex B Rotary Wing Aircraft Container Requirements.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC.

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N. W. Seventh Street, Miami, FL 33125.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A242 - High Strength Low-Alloy Structural Steel.
A606 - Steel Sheet and Strip, Hot Rolled and Cold-Rolled, High Strength, Low-Alloy, with Improved Corrosion Resistance.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The containers shall be noncollapsible of a permanent character and suitable for repeated use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

3.1.1 Terms and definitions. The terms used in this specification shall be as defined in ISO/1496/I.

3.1.2 Weight, ratings, and dimensions. The tare weight of the container shall be the minimum practical but shall not exceed 4,700 pounds for type I, 5,640 pounds for type II, 4,900 pounds for type III, and 5,840 pounds for type IV. The gross weight rating for each container shall be 20 long tons or 44,800 pounds. Dimensions, tolerances, and diagonal differences of the container shall meet the requirements for 20-foot nominal length containers as specified in ANSI MH5.4. All inside dimensions shall be the maximum attainable, consistent with design and material, but not less than those shown in table I. Door opening dimensions shall be in accordance with ANSI MH5.4.

TABLE I. Dimensions.

Item	Dimension
Inside width, minimum.	92 inches (type I & III); 91-1/2 inches (type II & IV).
Inside length, minimum.	231 inches.
Inside height, minimum (top to floor to underside of roof bows).	87 inches (type I & II); 93 inches (Type III & IV).

3.1.2.1 Tare weight. The tare weight to be marked on the container (see 3.6.1) shall be the average weight of the first 10 production units. The tare weight of the remaining production containers shall be within plus or minus 35 pounds of the stamped weight.

3.2 First article preproduction container. The contractor shall furnish two containers for examination and test within the time frame specified (see 6.2), to prove that his production methods will produce containers that comply with the requirements of this specification. Examination and tests shall be as specified in section 4, and unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified.

3.3 Materials. Materials shall be as specified herein. Materials not definitely specified shall be in accordance with a Federal, Military or National Technical Society, Association or Institute specification or standard. Dissimilar metals as defined in MIL-T-704 shall not be used together unless suitably protected against electrolytic corrosion.

3.3.1 Fiberglass reinforced plastic. The core material shall be laminated on both surfaces with 18 ounce per square yard (± 10 percent) woven roving fiberglass cloth reinforcement using thermosetting polyester resin. The surface laminates shall contain not less than 30 percent glass and a maximum of 70 percent polyester resin by weight. The woven roving cloth shall be continuous in both directions with no ties or laps. The exterior panel surface shall be coated with a gel coat fused under heat and pressure. The gel coat shall be ultra-violet, weather, mildew, oil and chemical resistant, shall conform to the color and finish requirements of 3.5.3.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:09Z
Check the source to verify that this is the current version before use.

3.3.2 Steel requirement. The container shall be constructed of high-strength low alloy (HSLA) structural and sheet steels conforming to ASTM A242, type I and ASTM A606, type 4. When specified, (see 6.2) material shall be corrosion resistant steel, modified type 409 or muffler grade stainless steel.

3.4 Construction. Containers shall be constructed so as to be free of any recesses and voids in which contraband can be concealed or where moisture can accumulate. No part of the container (when empty) shall protrude beyond the outside surfaces of the corner fittings. The yield point of materials shall not be exceeded when the container is tested as specified in 4.5.2. Material for sides, ends and top shall be steel or fiberglass reinforced plastic (FRP).

3.4.1 Side walls and front wall. The steel side walls and front wall may be of the interior or exterior post type, corrugated, or of smooth skin construction. On all types of construction, lap splices are permitted on metal skin. The interiors of metal side and front walls of type I containers, and the metal front walls of type II containers shall be lined, full height, with 1/4-inch thick plywood conforming to NN-P-530, group A, type I, grade 2 or group B, Grade B-C or C-C (plugged), exterior type. The better surface shall face inside the container. All plywood joints and ends shall be fastened to structural members. A kick plate 12 inches high and made of 18-gage metal or of 1/8 inch thick fibrous glass reinforced plastic laminate conforming to MIL-P-17549, grade W shall be attached to the bottom of the interior side and front end walls over the plywood liner. The kick plate shall be fully adhesive bonded to the plywood.

3.4.2 Floor construction. The container floor shall be of laminated or solid hardwood boards conforming to drawing 13219E0080. The boards shall run longitudinally and be of the same width and thickness, except that the width of the longitudinal edge boards may be different, if required. When the widths of both longitudinal edge boards differ from the intermediate plank width, they shall be of the same width as each other. Any joints in the floor boards shall conform to drawing 13219E0080. The board edges shall be jointed by shiplap construction. The boards shall be attached to the cross members by means of at least three fasteners per board, per crossmember, for boards over 7 inches wide; and two fasteners per board, per crossmember, for boards less than 7 inches wide. Countersunk head fasteners, 1/4 inch diameter minimum, either of the selftapping screw type, or machine screws with self-locking nuts, shall be installed so that each head is 0 to 1/16 inch below the board surface and at least one inch from the board edge. The floor shall be watertight. The floor shall be installed to permit lateral variations in floorboard width due to swelling. When specified (see 6.2) alternate flooring material such as preservative treated plywood or composite materials shall be provided.

3.4.3 Understructure. The container bottom side rails and bottom end frame members shall not deflect more than 1/4 inch below the bottom plane of the bottom corner fittings when the container is tested as specified in 4.5.2. Crossmembers shall not deflect below the bottom plane of the bottom corner fitting when the container is tested as specified in 4.5.2; except that when tested as specified in 4.5.2.2.8, they may deflect to not more than 1/4 inch below the bottom plane of the bottom corner fittings. All cross-members shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

be of the same configuration and strength and shall have a maximum center-to-center distance of 12 inches, except crossmember supports within 24 inches of the threshold shall have a minimum center-to-center distance of 8 inches. Load transfer zones shall be provided per ISO/1496/I.

3.4.4 Roof construction. The roof shall be self draining and of one piece construction. If the roof bows are fastened to the roof sheet they shall be bonded by a suitable bonding agent. In the case of roof bows that are not fastened to the roof sheet, antichafing material shall be affixed to the roof bows on the surfaces facing the roof sheets. The roof sheet, roof rails, and upper end frame members shall be at least 1/4 inch below the top plane of the top corner fittings.

3.4.5 Doors. Two doors shall be hung within the rear end frame and shall provide a clear opening conforming to the dimensions specified in 3.1.2. Heavy duty pin hinges, recessed within the corner structure, shall be provided on each door allowing the door to fold back against the sides of the body. Steel hinges shall have corrosion-resistant steel pins. Each door shall be provided with one or more heavy-duty, handle-operated cam locking devices with antirack provisions. Handles shall be 12 to 18 inches above the bottom plane of the bottom corner fittings. Each locking device handle must be capable to accept a padlock or security seal. Means shall be provided to hold the doors in the full open position. The doors when closed shall be sealed in such a manner as to prevent water entry into the container. All moving parts of the door locking mechanism and door hinges, if not permanently lubricated, shall be lubricated with graphite grease conforming to VV-G-671.

3.4.6 Corner fittings. The corner fittings shall meet the dimensional requirments of ISO/1161. The corner fitting shall be constructed of such materials as to enable them to pass the operating and testing requirements as specified herein.

3.4.7 Antipilferage provisions. Hinge-pins and screws, bolts, and other fasteners used for securing the hinges and closing devices to the container and for holding the essential parts of the sides, ends, and roof, shall be tack welded or otherwise secured in such a manner as to prevent access to the interior of the container without leaving visible signs of tampering. Fasteners used for FRP panels will be of sealed head type not allowing removal on the exterior of the container. Where such welding destroys protective coating on the items being welded or on other container parts, the weld and surrounding area shall be thoroughly cleaned, treated, and painted in accordance with 3.5. All locking device handles shall be furnished with provisions for padlocking and customs sealing.

3.4.8 Restraint system. The internal cargo restraint system in each type II and IV container shall be of a make and model capable of accepting the beam shoring per drawing 13219E0039 which has been approved for use in the shipment of ammunition in containers by the Bureau of Explosives of the AAR, the U.S. Coast Guard, and the US Army Defense Ammunition Center and School. If a belt rail system is selected, it shall consist of eight side rails attatched to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

walls of the container, a vertical rail attached on each side of door frame, and 25 cross rails. The lower rails on the side walls shall be located so that the top surfaces of cross rails, when attached to side rails, will be 4, 16, 28, 38, 48, 60 and 72 inches respectively above the floor. The uppermost side rails shall be located as high as possible without interfering with the upper roof rails of the container. The vertical rails in the door frame shall provide for location of cross rails at approximately the same heights as provided by the side rails. Side rails shall extend continuously from the front wall to the rear doors of the container.

3.5 Cleaning, treatment, and painting. Cleaning, treatment, and painting shall be in accordance with MIL-T-704. Ferrous metal surfaces shall conform to type F, all non-ferrous surfaces, aluminum, FRP, etc., shall conform to type G. After cleaning, treatment, and primer, all exposed interior and exterior surfaces shall be painted with polyurethane conforming to MIL-C-46168, color forest green. Painting of interior metal surfaces covered with plywood is not required. Crossrails when used in the mechanical restraint system shall not be painted.

3.5.1 Wood surfaces. All wood surfaces shall be cleaned of all dirt, grease, oil, and other deleterious substances before preservation treatment. Painting of the exterior wood surface shall be as specified for metal surfaces.

3.5.1.1 Plywood. Exterior plywood surfaces shall be treated prior to painting in accordance with MIL-T-704. Treatment of interior plywood members is not required.

3.5.1.2 Other wood members. Unless otherwise specified herein, other wood members shall be preservative treated prior to installation and painting in accordance with MIL-STD-171, finish number 25.1, or the immersion method of finish number 25.3.

3.5.2 Undercoating. The underside of the container floor shall be coated after painting with a 1/16-inch bituminous coating conforming to MIL-C-83933. Overrun onto the crossmembers, corner fittings, side rails, and end frame members is permitted.

3.5.3 Plastics. Plastics used in exterior applications shall be pigmented and finished to produce a color, forest green, to match with that of the steel frame, or paint all plastic surfaces in accordance with 3.5.

3.6 Identification marking. All letters shall be Gothic or futura capitals with Arabic numerals and shall read from left to right.

3.6.1 Exterior marking. The container shall be identified and marked in accordance with ANSI MH5.3 with the following additional requirements:

a. The following data shall be marked on the right-hand door in characters not less than 2 inches high:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

(1) Maximum gross weight in pounds.
(2) Tare weight in pounds.
(3) Net cargo weight in kilograms, tons, and pounds.
(4) Shipping cube in cubic meters and cubic feet.
(5) Nominal outside dimensions in meters and feet.

b. The owner code and the serial number shall be as specified (see 6.2). All markings shall be white stencil paint conforming to TT-P-98, type I. Spacing between letters shall be even to present a neat, readable appearance. Containers with external corrugated or post construction that prohibits marking entirely on smooth surfaces shall be equipped with metal marking plates. Marking plates shall be attached to metal containers by not less than eight fillet welds each not less than 1 inch in length; or by not less than 12 spot welds evenly distributed.

c. The following warning shall be marked on both sides at the base line, centrally located, "Caution do not handle container with fork lift tines", letter size to be minimum 3 inches.

3.6.2 Interior marking. The container control number shall be stamped or welded in characters not less than 1/2 inch high on an interior surface of the rear upper end frame member. The number shall be located within an area 18 inches from the left vertical end frame member where it will not be obscured.

3.6.2.1 Instructions for stowage of crossmembers, type II and type IV. Each type II and type IV container, equipped with a belt rail cargo restraint system, shall be stenciled on the inside surface of the right hand door with white letters 1-1/4 inches high as follows: "25 crossbeam assemblies shall remain with the container. Unused beams will be mounted in belt rails in loaded and empty containers."

3.6.3 Data plates. A nonferrous metal data plate conforming to MIL-P-514, type I, style 3, composition C shall be affixed to the center of the external surface of the right hand door. The method of attachment shall be by either riveting or bolting. The data plate shall contain the following information:

(NAME OF OWNER, see 6.2)
CONTAINER, CARGO
SPECIFICATION: MIl-C-52661
FSN:_____
CONTROL NUMBER:_____
TARE WEIGHT: LBS_____KG_____
CONTRACT NUMBER:_____
MFD BY:_____
DATE (MONTH & YEAR):_____
TECHNICAL MANUAL:_____

3.6.4 CSC plate. A Convention for Safe Containers (CSC) plate shall be applied for and obtained from a designated approval authority, attached and displayed as required by the convention in accordance with CFR 49, parts 450 and

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:09Z
Check the source to verify that this is the current version before use.

451. The CSC plate shall be attached to the right door of the container approximately 6 inches from corner post midway between middle and bottom hinge.

3.7 Document holders. One waterproof, thru-door metal documentation box shall be securely fastened to the right door of each container. The closing and latching devices of the boxes shall be capable of repeated opening and closing, without tools and without impairing their sealing quality. The inside dimensions of the box shall be not less than 9 x 12 x 2 inches. Each box shall be stenciled "Document Holder" in white lettering 1 inch high on both sides. A padlocking device shall be provided on the outside of the box.

3.7.1 Placard holder. Four stainless steel placard holders conforming to DOT, BOE-600-C, part 172, appendix C, shall be provided, one each located at lower center of each side; one each at upper left quadrant of the closed end, and one each lower left quandrant of the door end. Placard holders shall be permanently attached, welded when applicable, in the designated locations.

3.8 Performance. Each container, when carrying its maximum permissible load, shall be capable of fulfilling the following operating requirements, and shall be designed and fabricated to withstand the test requirements of 4.5.2.

3.8.1 Stacking. Stacking capability shall be in accordance with ISO/1496/I.

3.8.2 Lifting from the top. Lifting from the top capability shall be in accordance with ISO/1496/VII, Annex B, Rotary wing.

3.8.3 Lifting from the bottom. Lifting from the bottom capability shall be in accordance with ISO/1496/VII, Annex B, Rotary wing.

3.8.4 Longitudinal restraint. Restraint capability shall be in accordance with ISO/1496/I.

3.8.5 Floor. The floor of the container shall withstand loading in accordance with ISO/1496/I.

3.8.6 Roof. The container roof shall withstand loading in accordance with ISO/1496/I.

3.8.7 Walls. Each of the end and side walls shall withstand loading in accordance with ISO/1496/VII (Rotary Wing), with the additional requirement that each side wall of a type II and IV container (when equipped with a belt rail cargo restraint system) shall withstand the full maximum payload for a 20-foot container.

3.8.8 Transverse rigidity. The container shall withstand racking loads in accordance with ISO/1496/I with the additional requirement that the door end of the container shall show no signs of water leakage during the tests specified herein.

3.8.9 Weatherproofing. The container shall show no signs of leakage when tested in accordance with 4.5.2.2.9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

3.8.10 Railroad car impact. The container shall show no sign of damage or permanent deformation which would render it unsuitable for its intended use following each hump test.

3.9 Workmanship. All parts, components, and assemblies of the container including castings, forgings, molded parts, stampings, seals and sealing agents, machined surfaces, and welded parts shall be clean and free from any defects that will reduce the capability of the container to meet the requirements specified herein. Any components and assemblies which have been repaired or modified to overcome deficiencies shall not be used without prior specific approval of the contracting officer. External surfaces shall be free from burrs, slag, sharp edges, and corners except where sharp edges and corners are required. The internal cargo space shall be free from sharp protrusions that could damage cargo.

3.9.1 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and clipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting, using a tip suitable for the thickness of the metal, may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be free of slag. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.9.2 Fiberglass. Fiberglass or fiberglass coated plywood used in the fabrication of the container shall be smooth and free from bubbles, glazing cracks, and discontinuities. All edges of fiberglass reinforced plywood shall be completely sealed to prevent water intrusion between the laminates.

3.9.3 Bolted and riveted connections. Bolt and rivet holes shall be accurately punched or drilled and shall have the burrs removed. Washers, lockwashers, or lock nuts shall be provided where necessary and all bolts, nuts, and screws shall be tight. Rivet heads, when not countersunk or flattened, shall be of uniform size and shape for the same diameter of rivet, concentric with the rivet holes, and in full contact with the surface of the members.

3.9.4 Welders and welding.

3.9.4.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedures of the American Welding Society.
Welding Qualifications of the ASME.

3.9.4.2 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Weld penetration shall be such

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration.

3.9.5 Machined work. All parts shall be manufactured to gage through the use of jigs and fixtures and shall provide interchangeability of parts as manufactured.

3.9.6 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or defects which might render the castings unsound for use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specifid herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements are classified as follows:

a. First article preproduction inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First-article preproduction inspection.

4.3.1 In-process inspection. The two first-produced containers shall receive an in-process inspection by the contractor under the direction and surveillance of the Government. This inspection shall be conducted to evaluate materials and workmanship as specified in 3.3 and 3.9. Nonconformance to 3.3 or 3.9 shall be cause for rejection.

4.3.2 Examination. The two first-produced containers shall be examined as specified in 4.4.1 and 4.5.1. Presence of one or more defects in either container shall be cause for rejection of both containers.

4.3.3 Tests. The two first-produced containers shall be tested as specified in 4.5.2 through 4.5.2.2.11. Tests specified for single containers shall be performed on both containers. Failure of any test shall be cause for rejection

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

of both containers and shall be cause for the Government to defer future acceptance until objective evidence furnished by the contractor indicates that deficiencies revealed by the tests have been corrected.

4.4 Quality conformance inspection.

4.4.1 Examination. Each container shall be examined for defects listed in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each container shall be tested for weatherproofness in accordance with 4.5.2.2.9 before application of finish painting or undercoating, or installation of a plywood liner. Penetration of moisture shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The containers shall be examinmed for the following defects:

Major

101. Dimensions not as specified. Each incorrect dimension shall be considered one defect.
102. Components missing or not as specified. Each missing component shall be considered one defect.
103. Assembly incomplete or not as specified.
104. Doors do not open, close, or latch when operated by one man standing on the ground and without the use of tools.
105. Cleaning, treatment, or painting not as specified.
106. Door seals not as specified.
107. Tare weight in excess of that specified (see 3.1.2).
108. Tare weight not within specified limits (see 3.1.2.1).

Minor

201. Door hinges and latches not lubricated as specified.
202. Workmanship not as specified.
203. Fasteners missing.
204. Fasteners not countersunk as specified.
205. Antipilferage provisions not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. The container shall first be tested for weather-proofness before application of finish painting or undercoating or installation of plywood liner. Other tests shall be made after application of finish painting and undercoating and installation of plywood liner (if required). A second test for weatherproofness shall be made on the doors during racking tests of the door end. A final test for weatherproofness shall be made after completion of all other tests. The test load within the container shall be uniformly distributed, except as otherwise specified herein. On completion of each test, the container

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

shall be examined to determine compliance with the performance requirements specified in 3.8.

4.5.2.2 Test procedures.

4.5.2.2.1 Stacking. A stacking test shall be conducted in accordance with ISO/1496/I.

4.5.2.2.2 Lifting from the top. Top lifting capability shall be tested in accordance with ISO/1496/VII Annex B, Rotary wing.

4.5.2.2.3 Lifting from the bottom. Bottom lifting capability shall be tested in accordance with ISO/1496/VII Annex B, Rotary wing, with the additional requirement that the line of action of the lifting cables and the outer face of the corner fittings shall be parallel and not more than 1-1/2 inches apart.

4.5.2.2.4 Longitudinal restraint. Restraint capability shall be tested in accordance with ISO/1496/I, with the additional requirements that each side may be tested individually using a force of 56,000 pounds in tension and compression.

4.5.2.2.5 End wall strength. A 20-foot container shall be loaded with a weight equal to 0.83 times the maximum container payload (44,800 pounds less the tare weight of the container). The container, with the end to be tested facing down, shall be supported on two level pads placed under the end faces of the two bottom corner fittings. The container shall be held in this position by restraining devices attached to the bottom corner fittings at the opposite end. The supporting pads shall be of sufficient thickness to allow the ends of the container to cantilever freely in space when loaded as specified herein. Structural members and panels not under test may be supported during this test. The load shall be applied to the full cross section of the end wall, and shall be uniformly distributed. Tests shall be made on both the front end and on the rear (door) end. The load shall remain in place for not less than 5 minutes. A test on the doors and door frame not attached to a container is not an acceptable substitute for the specified test on the door end.

4.5.2.2.6 Side wall strength. The container shall have one side wall tested when both sides are identical. If the two side walls are not the same, both shall be tested. The container shall be loaded to 0.83 times its maximum payload for type I and III containers and to its maximum payload for type II and IV containers. Water shall be used for the test load. The container, with the side to be tested facing down, shall be supported on two level pads placed under the side faces of the two bottom corner fittings. The container shall be held in this position by restraining devices attached to the bottom corner fittings at the opposite side. The supporting pads shall be of sufficient thickness to allow the sides of the container to cantilever freely in space when loaded as specified. The load shall be applied to the full cross section of the side wall and shall be uniformly distributed. The roof may be supported during this test. Load shall remain in place for not less than 5 minutes.

4.5.2.2.7 Roof strength. Roof strength shall be tested in accordance with ISO/1496/I with the additional requirement that test loading shall be applied for not less than 5 minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

4.5.2.2.8 Floor strength. Floor strength shall be tested in accordance with ISO/1496/I with the additional requirement that static test loads shall be applied for not less than 5 minutes. Tests shall be made with the container resting on its four bottom corner fittings, with the intermediate structure not less than 1 inch above the supporting surface and free to deflect.

4.5.2.2.9 Weatherproofness. Each container shall be tested for weatherproofness before application of finish painting or undercoating or installation of the plywood liner. A stream of water shall be applied to all joints, seams and gasketed closures (except during racking tests) of the container from an 0.5 inch inside diameter nozzle at a nozzle pressure of not less than 15 psig. Rate of movement over the exterior of the container shall be approximately 4 feet per second. The nozzle shall be held 5 feet away from the container under test. Tests shall be continued for not less than 15 minutes (except during rigidity tests). On completion of the test the container shall be examined for penetration of water into the interior of the container. Penetration of moisture shall constitute failure of this test.

4.5.2.2.10 Transverse and longitudinal rigidity. Test shall be conducted in accordance with ISO/1496/I with additional requirements that, (1) both doors shall be closed and latched during test, (2) racking forces shall be applied first laterally and then longitudinally to each top corner fitting in turn, (3) each force application shall be for not less than 5 minutes, and (4) the container shall be tested for weatherproofness during each force application to a top rear corner fitting. The weatherproofness test shall conform to 4.5.2.2.9, except that the water shall be applied to the door end of the container for not less than 5 minutes.

4.5.2.2.11 Railcar impact test. The railcars shall be AAR approved for domestic rail freight service. The test car (hammer car) shall be AAR approved for container on flatcar (COFC) service. The buffer cars shall have no cushioning devices other than standard pocket draft gears. The container shall be secured to the test car using the test car's AAR approved locking fittings. The container shall be fully loaded for the conducting of the test. Container types II and IV shall be fully loaded using dummy loads equal in weight and general character to class 'A' explosives. The load within the container shall be secured using the mechanical restraint devices. Hump the test car, with the container tied down, at speeds of not less than 4, 6, and 8 mph respectively against 5 empty freight cars with their brakes set and draft gear extended. The container shall face in the direction of the 5 cars. Repeat the procedure at not less than 8 mph with the door end of the container in the direction of the 5 cars. The container may be reversed by either reversing the direction of the flatcar (preferred) or by reloading the container in the proper direction on the freight car. Once the test has begun, there shall be no readjustment of load, nor any reconditioning of the tiedowns except as necessary to reverse the direction of the container. After each impact, the container shall be inspected. After completion of the railroad impact test, an operational checkout of the container shall be conducted. Nonconformance to 3.8.10 shall constitute failure of this test.

4.5.2.2.12 CSC approval testing. Additional tests required to obtain convention for safe containers (CSC) approval. Testing as required to conform to the requirements of 3.6.4 and 3.8.1 thru 3.8.10.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

4.5.2.3 Failure criteria. Any of the following shall constitute failure of any of the tests specified in 4.5.2. After removal of the load, and before checking criteria of "d", "h", and "j", the container under test shall be subjected to shock and vibration (ISO/1496/I) to eliminate temporary residual bulge.

a. Deflection of the container bottom side rails or bottom end frame members of more than 1/4 inch below the bottom plane of the bottom corner fittings.
b. Deflection of the container crossmembers below the limits specified in 3.4.3.
c. Evidence of weld failure.
d. Permanent deformation (deformation which persists for 2 hours after removal of the load and subjecting the container to shock and vibration as defined in ISO/1496/I shall be considered permanent).
e. Any broken or torn component.
f. Any cracked or broken floorboard.
g. Any detachment of a gasket.
h. Inability of one man standing on the ground to open or close a door without the use of tools.
i. Inability of one man standing on the ground to operate a latching mechanism without the use of tools.
j. Any dimension after a test not within the specified tolerances (dimension changes which persist for 2 hours after removal of the load and subjecting the container to shock and vibration as defined in ISO/1496/I shall be considered permanent).
k. Penetration of water into the interior of the container.
l. Inability to obtain CSC approval.

4.6 Inspection comparison. The Government may select containers at any time during the contract production period and subject these containers to the examination specified in 4.5.1 and the test specified in 4.5.2 (except for the first weatherproofness test specified in 4.5.2.1) to determine conformance to the requirements of this specification. This inspection will be performed by the Government at a site selected by the Government on containers selected at random from those which have been accepted by the Government and will not include the previously inspected first-produced containers. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as a part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison container to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison container and shall be cause for refusal by the Government to continue acceptance of production containers until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on containers previsously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

containers accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging.

4.7.1 Quality conformance inspection of pack.

4.7.1.1 Unit of product. For the purpose of inspection, a container prepared for shipment shall constitute a unit of product.

4.7.1.2 Examination. Each container shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

109. Markings missing, illegible, or incorrect.

5. PACKAGING

5.1 Preservation and packing. Preservation and packing is not applicable for ISO cargo containers.

5.2 Marking. Destination markings for shipment from the contractor's plant to the initial receiving activity shall be marked on tags with waterproof ink and the tags securely fastened to the container door handle hardware. Other required contract markings can be found on the container data plate.

6. NOTES

6.1 Intended use. The reusable cargo container is intended for transporting military and quasi-military cargo. The container can be lifted from the top or bottom corner fittings, and can be carried by highway, railroad, water, and air modes of transportation.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type of container required (see 1.2).
c. Date of issue of DoDISS applicable and exception thereto (see 2.1.1).
d. When the Government will conduct any or all of the first-article preproduction container examination and test. When the Government will conduct a portion of the first-article pre-production examination and tests, the contracting officer should specify which examination and test will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
e. Time frame required for submission of first-produced containers (see 3.2).
f. When corrosion resistant steel is required (see 3.3.2).
g. When alternate flooring material is required (see 3.4.2).
h. Owners code and serial number for each container (see 3.6.1).
i. Name of owner to be inscribed on data plate (see 3.6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

6.3 First article preproduction containers. Any changes or deviation of production containers from the approved first article preproduction containers during production will be subject to the approval of the contracting officer. Approval of the preproduced containers will not relieve the contractor of his obligation to furnish containers conforming to this specification.

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activities:
Army - MT, GL

Project 8115-A461

User activity:
Army - SM

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for us submitting comments and suggestions for improvements. All users of military standardization documents are invited to pro suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorizatio to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTA
NECESSA
IF MAILE
IN THE
UNITED STA

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Mobility Equipment R&D Command
ATTN: DRDME-DS
Ft. Belvoir, VA 22060

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-52661B(ME)	Containers, Cargo

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:09Z
Check the source to verify that this is the current version before use.